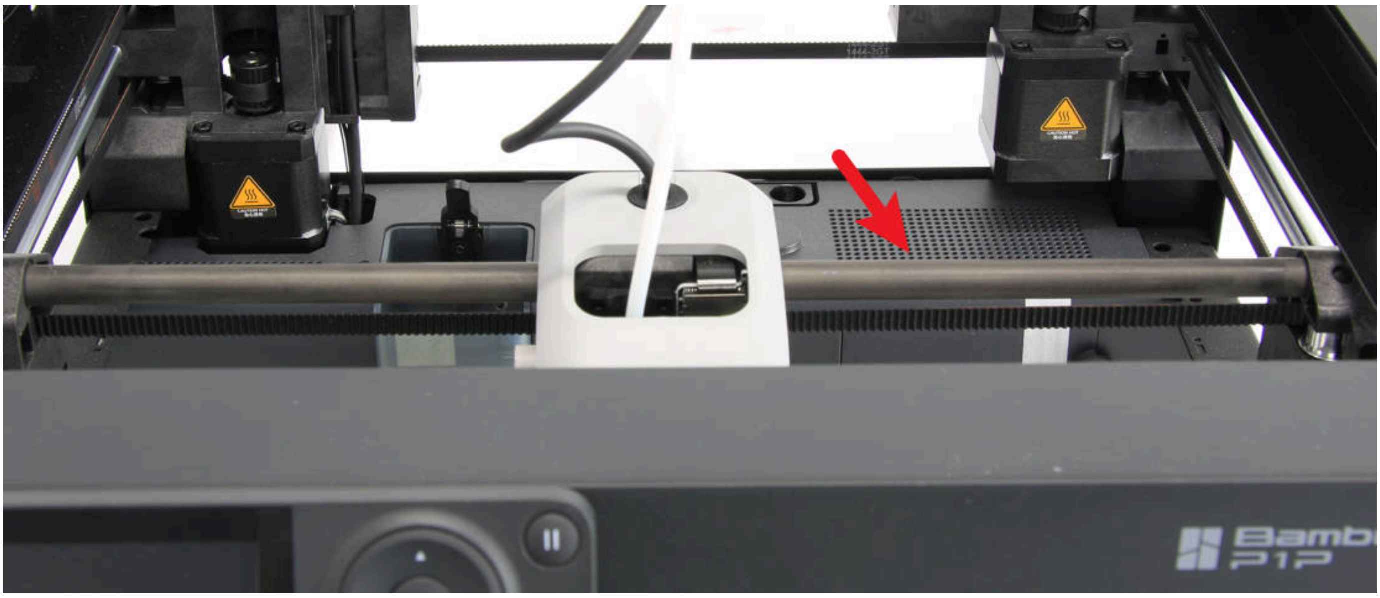
Bambu Lab P1 series printer Maintenance Recommendation

Regular maintenance can minimize wear and extend the printer's lifespan while also lowering the risk of a print failure. Please visit the Spare Parts Store Page to purchase spare parts and other maintenance items (such as grease or lubricating oil) referred to below. Additionally, some consumables are available in a convenient bundle here .

X-axis Carbon Rods

While the Carbon rods used on the X-axis do not use lubrication, it's still recommended to clean them periodically due to dust and other buildup.

WARNING: Do not use grease on the carbon rods as it will cause additional buildup and premature wear or part failure.



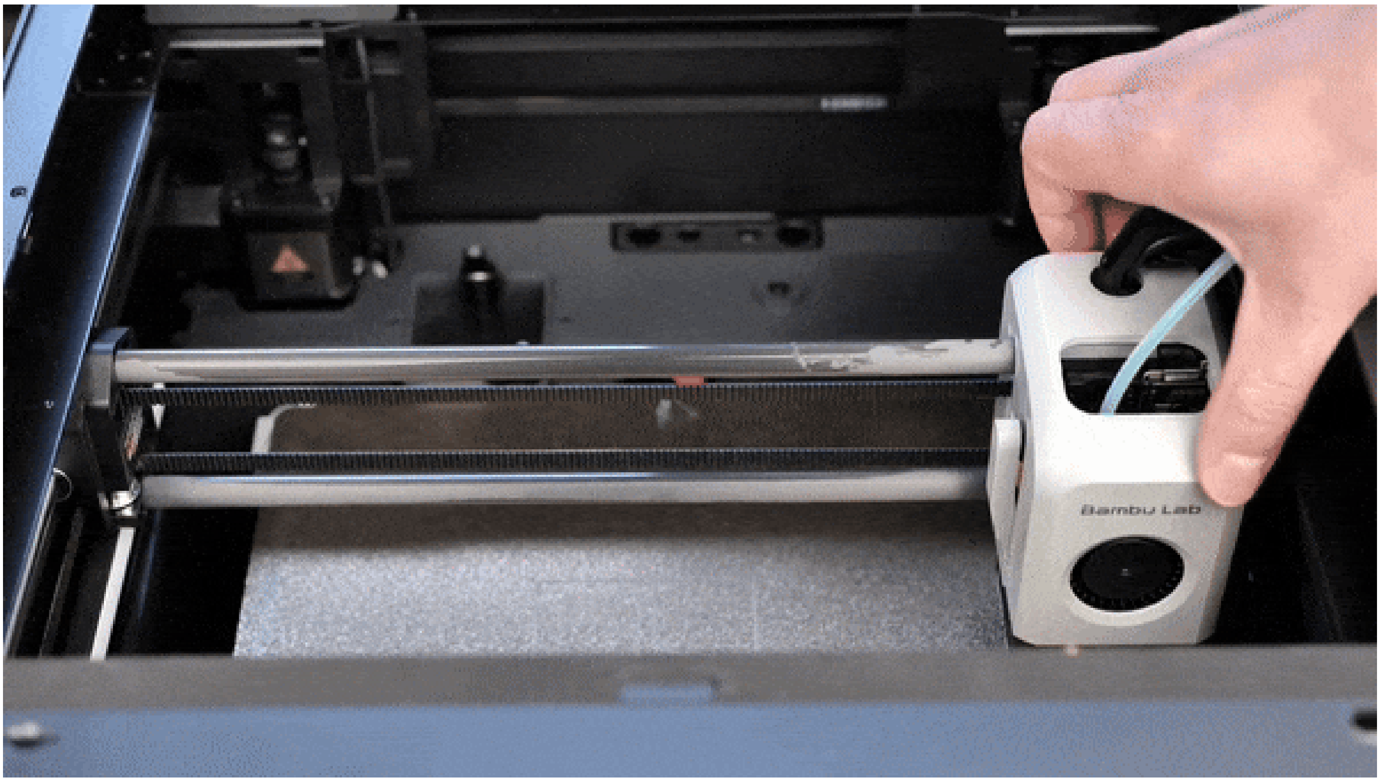
When to do it

The x-axis carbon rods should be checked once a month for any dust and particle buildup.

How to do it

Apply a generous amount of **IPA (isopropyl alcohol)** to a clean microfiber cloth or paper towel, or directly to the carbon rods. Then, gently wipe along the length of each rod, removing any dust or buildup. It is recommended to repeat this step 2-3 times, until there is no more debris on the cleaning cloth. Be sure to move the print head to clean the entire length of both carbon rods.

It is also recommended to move the print head along the X axis a few times, with the carbon rods soaked with IPA, to help clean the inside of the bushings.



Once cleaning is complete, move the X axis back and forward along the Y axis a few times, then push it towards the back of the printer.

A video guide showing the same procedure is available here:

Cleaning Guide for Carbon Rods on a Bambu Lab F

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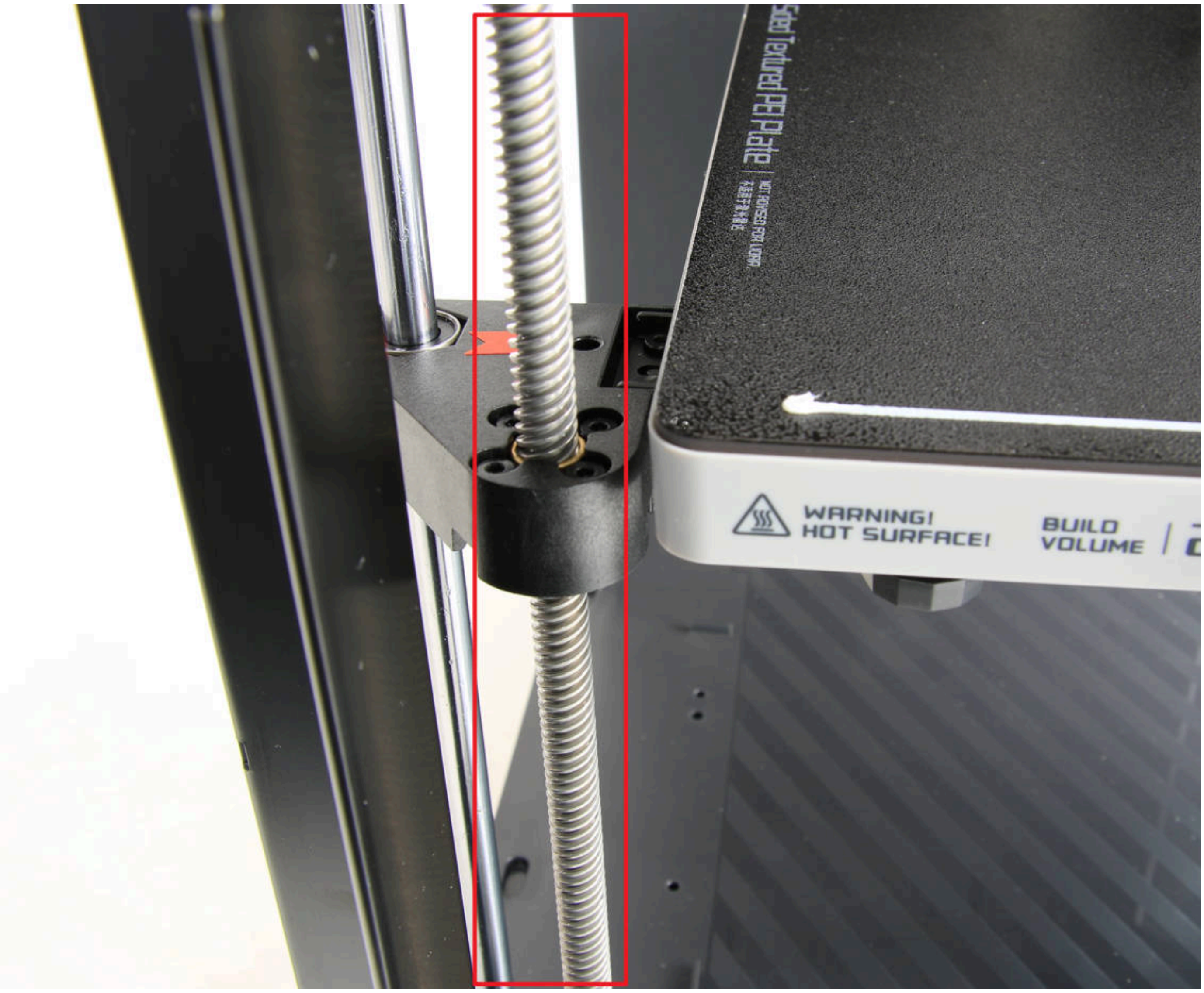
Z-axis Lead Screws

Three Z-axis lead screws require regular greasing. They are used for moving the heated bed on the Z axis, and proper greasing will ensure smooth operation.



When to do it

The z-axis lead screws should be checked and greased every three months.

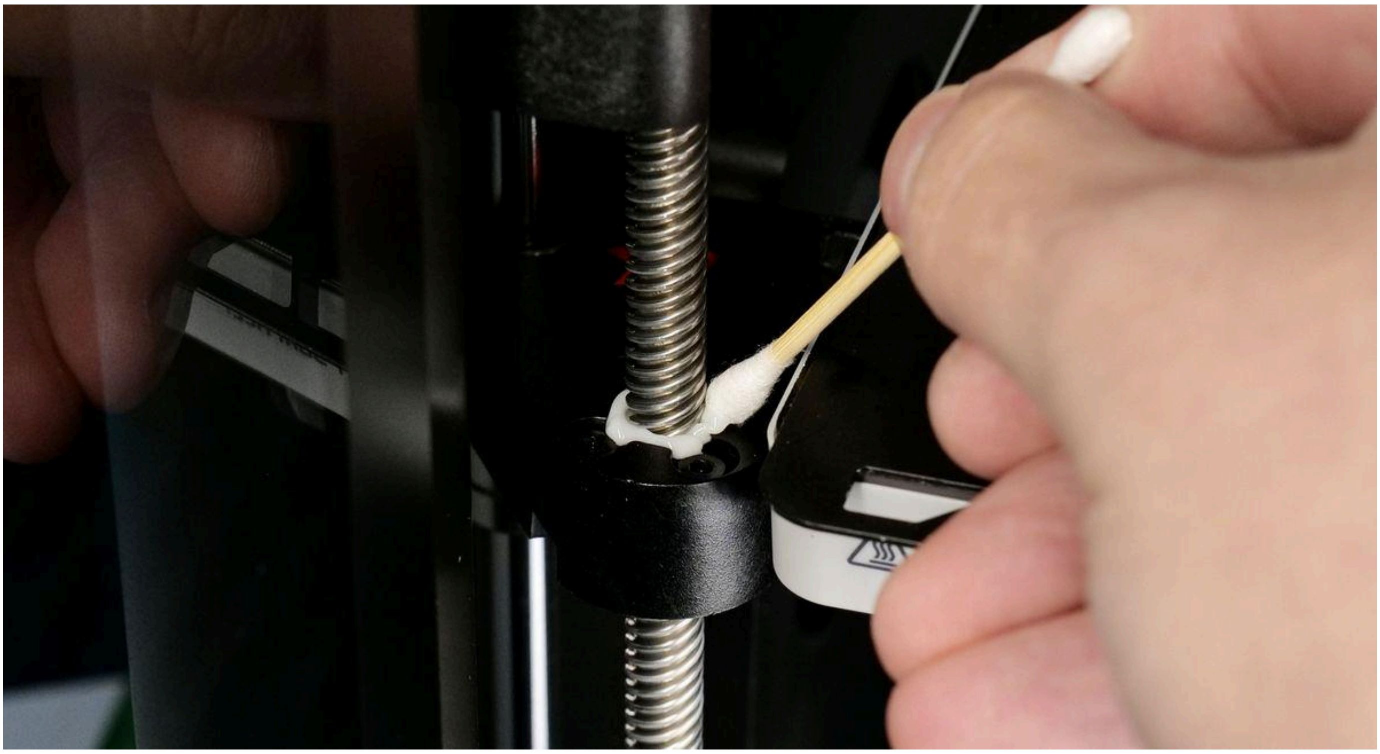
How to do it

Before greasing the z-axis lead screws, the first step before would be to clean them of any dust or plastic particles.

The next step is to use lubricating grease and apply a thin coat over the lead screws. With the bed home, apply a thin coat of lubricating grease, then move the bed to a lower position.

Apply another thin coat of grease on the z-axis lead screws and home the printer again.

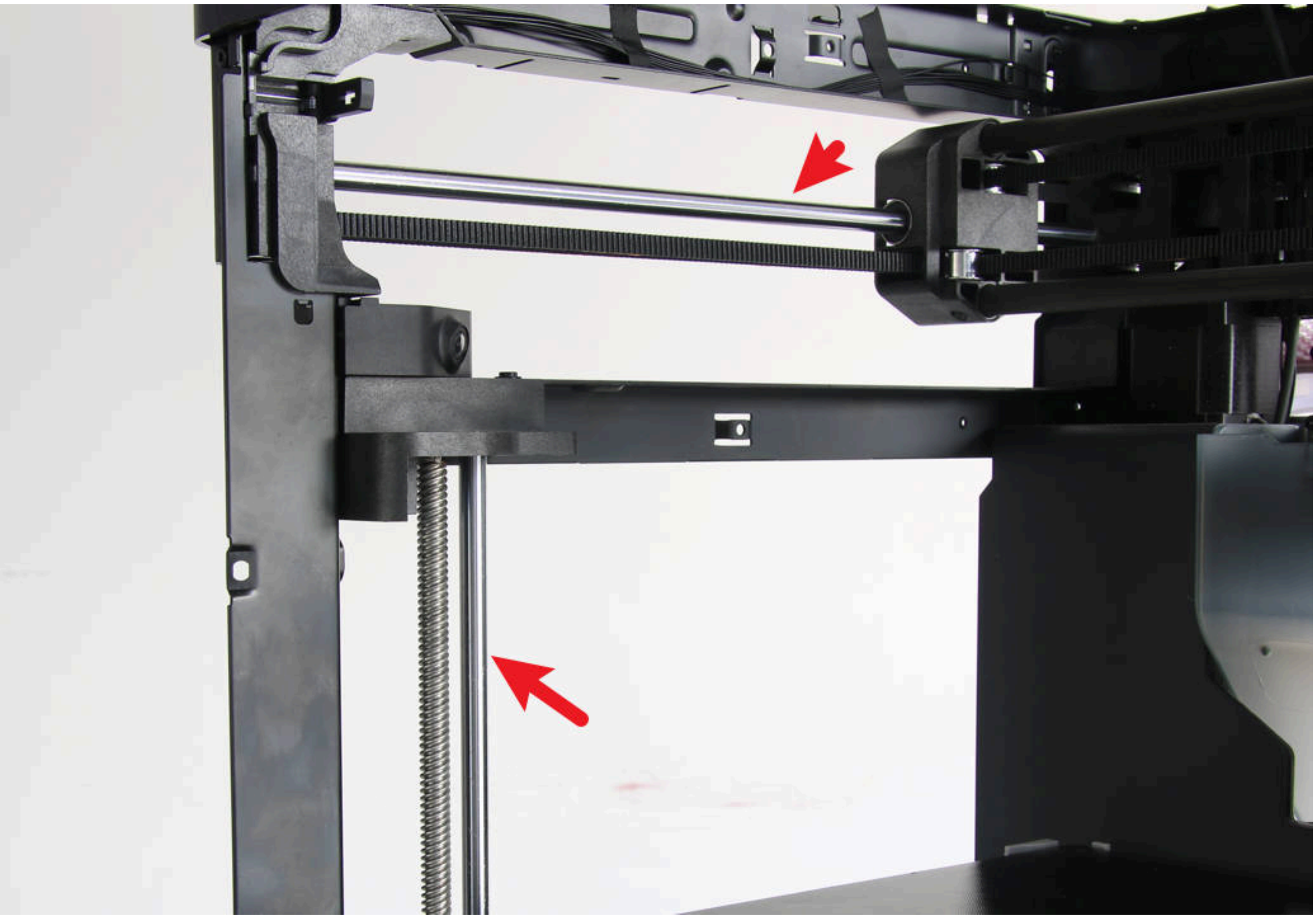
You can repeat the movement process a few times to ensure the grease is evenly spread over the z-axis lead screws. When complete, clean out any excess grease that builds up close to the lead screw nuts.



Please also refer to the How to lubricate the lead screws. You can use the BX-300/F series grease in the link, but other lubrication alternatives can be used. A popular lubricant is Super Lube 92003 Silicone Lubricating Grease with PTFE or Lucas Oil 10533 White Lithium Grease which should work similarly.

Y-axis and Z-axis Linear Rods and Bearings

To ensure smooth movement, P1 printers use LMU8 bearings and 8mm metal linear rods. The bearings are greased from the factory in order to prevent rust and prolong the service life. Regular cleaning and anti-rust maintenance are recommended to keep them functioning well indefinitely.



When to do it?

Y-axis and Z-axis linear rods should be checked once a month for any dust and particle buildup. If volatile filaments such as ABS\ASA are used, it is recommended to clean the rods every 5 rolls.

Y-axis and Z-axis rods should also have an anti-rust treatment every three months.

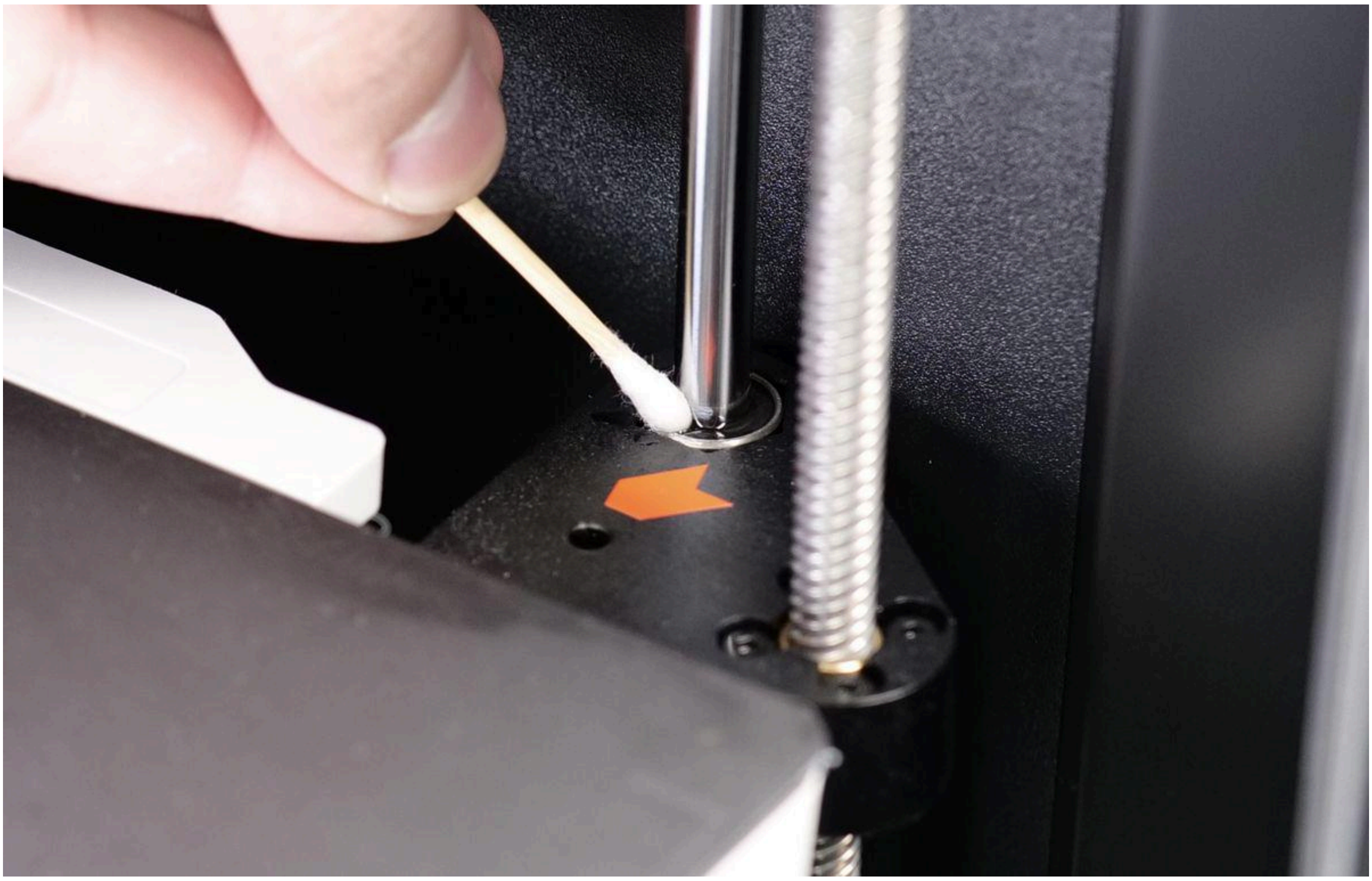
How to do it

Y-axis and Z-axis linear rods can be cleaned with isopropyl alcohol and a dust-free cloth. Spray a bit of isopropyl alcohol on the cloth and gently rub the rods to remove any debris.

Please note that if you clean the rods with IPA, you need to add a uniform thin coat of oil/grease on the rods, to prevent rust. Besides that, ensure that IPA does not reach the bearings of the X and Y axis.

If the bearing has abnormal ball noise due to insufficient internal lubrication, you can also try to apply some grease on the bearing and slide the bearing several times to make the grease soak into the bearing to reduce the abnormal noise. This may be challenging, so since the abnormal sound will not affect the printing performance in theory, it is not mandatory.

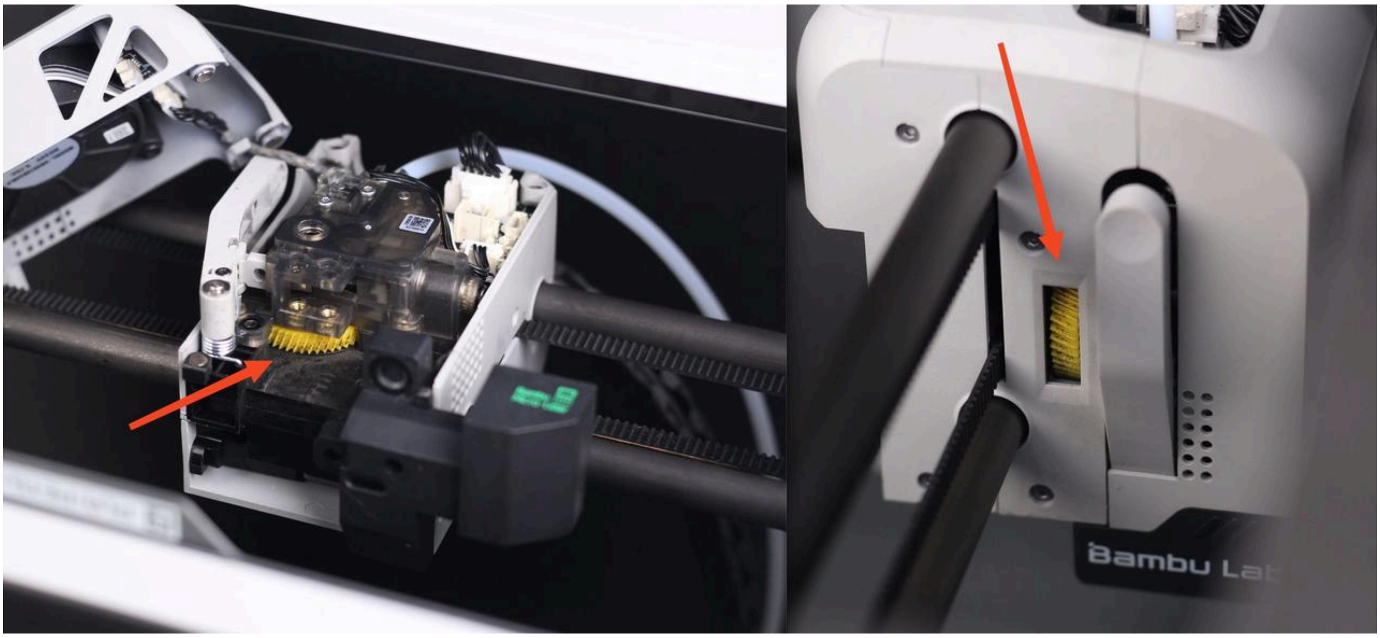
Avoid applying too much oil as it can drip on the linear bearing. If that happens, use a q-tip to remove any excess oil.



You can refer to using Super Lube 52004 Synthetic Lightweight Oil for better performance.

Extruder Assembly

The P1P Extruder assembly can get filament dust and debris inside after using the printer for a long time. Some filaments generate more dust than others, so your mileage may vary.



When to do it

You should clean the dust inside the extruder when you can see small amounts of dust on the yellow gear.

How to do it

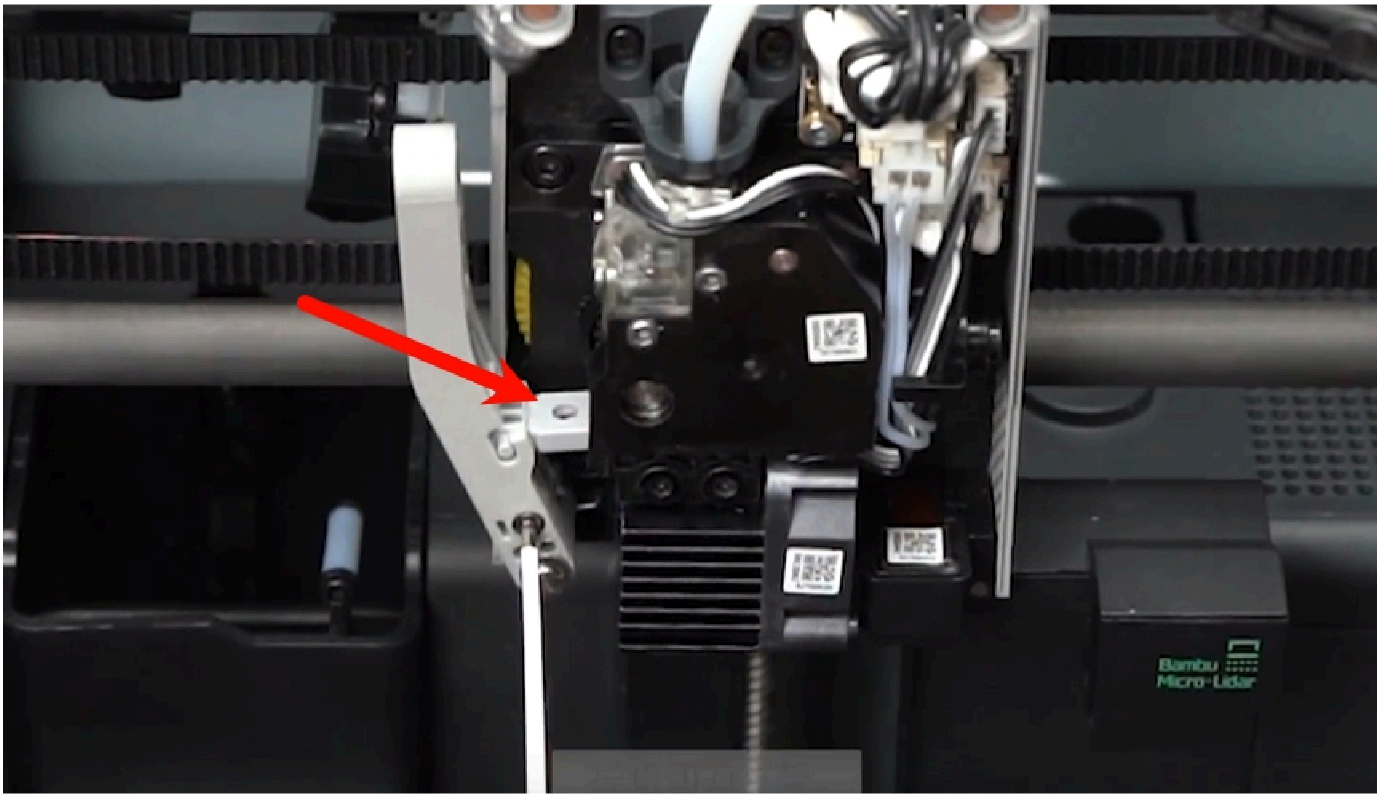
The fastest way to clean it and ensure the extruder is free from debris or filament dust is to use a can of compressed air and blow air over the yellow gear.

You can also remove the hotend and blow some compressed air underneath the extruder. This should clean most of the filament dust inside.

We also recommend checking the Extruder Maintenance Guide: Cleaning and Lubrication Wiki article for detailed information on how to clean the extruder.

Filament Cutter

The filament cutter used in the extruder cuts the filament regularly, during filament swaps. The blade of the cutter can get dull after a few rolls of filament are printed so it should be checked regularly to ensure that the blade is still sharp. Replacement blades may be purchased here .



When to do it

For regular filaments like PLA/PETG/ABS/PC, the blade should be checked every 3-5 rolls. If the blade is dull, replace it.

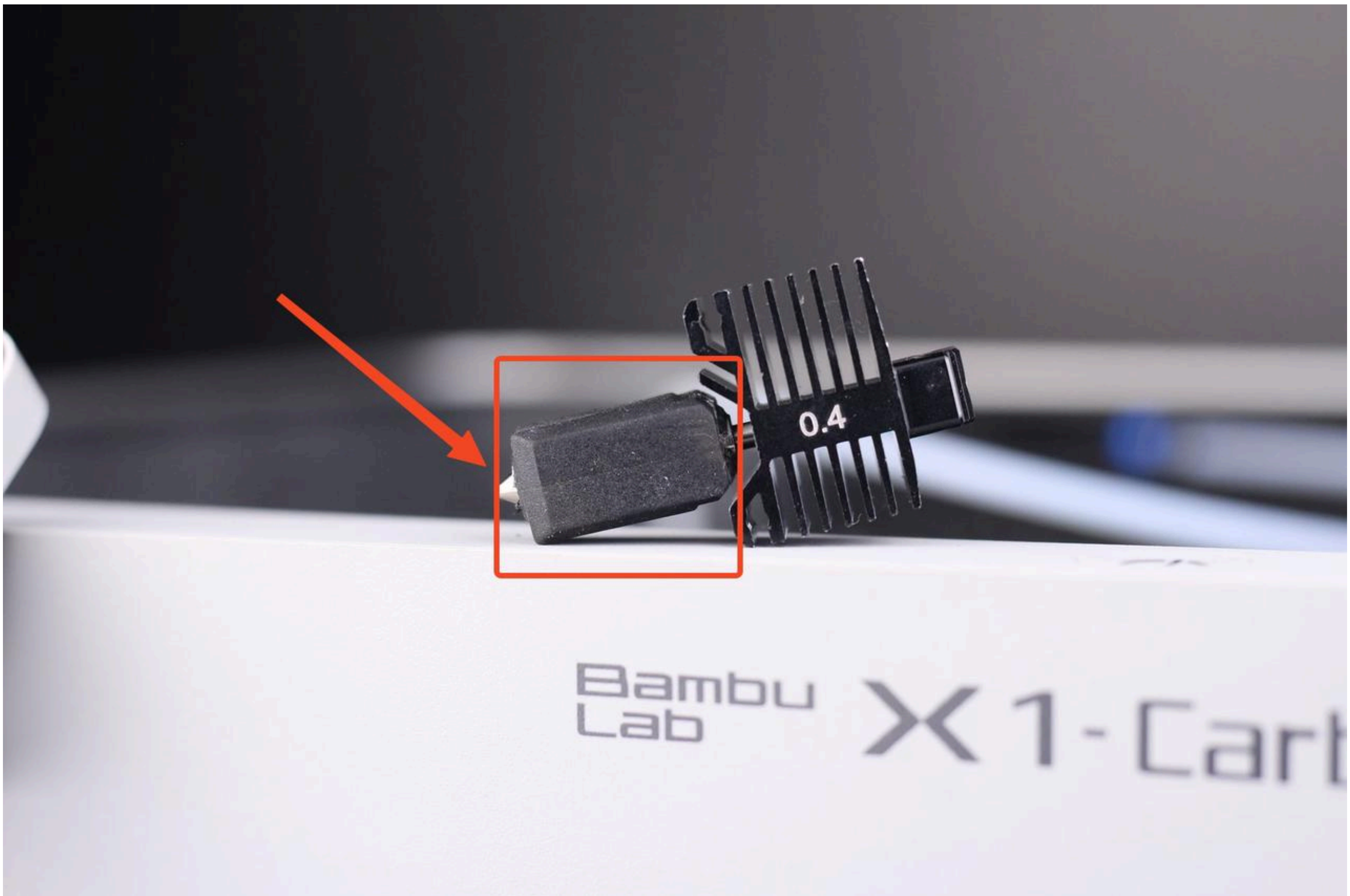
For abrasive filaments like PA+CF/PA+GF, the cutter blade can get dull much quicker, so we recommend checking it after printing 1-2 rolls of abrasive materials. If the blade is dull, replace it.

How to do it

The process of replacing the filament cutter is described in this Wiki article.

Hotend Silicone Sock

The silicone sock around the hotend helps maintain a consistent temperature while also protecting the hotend from plastic buildup during printing.



When to do it

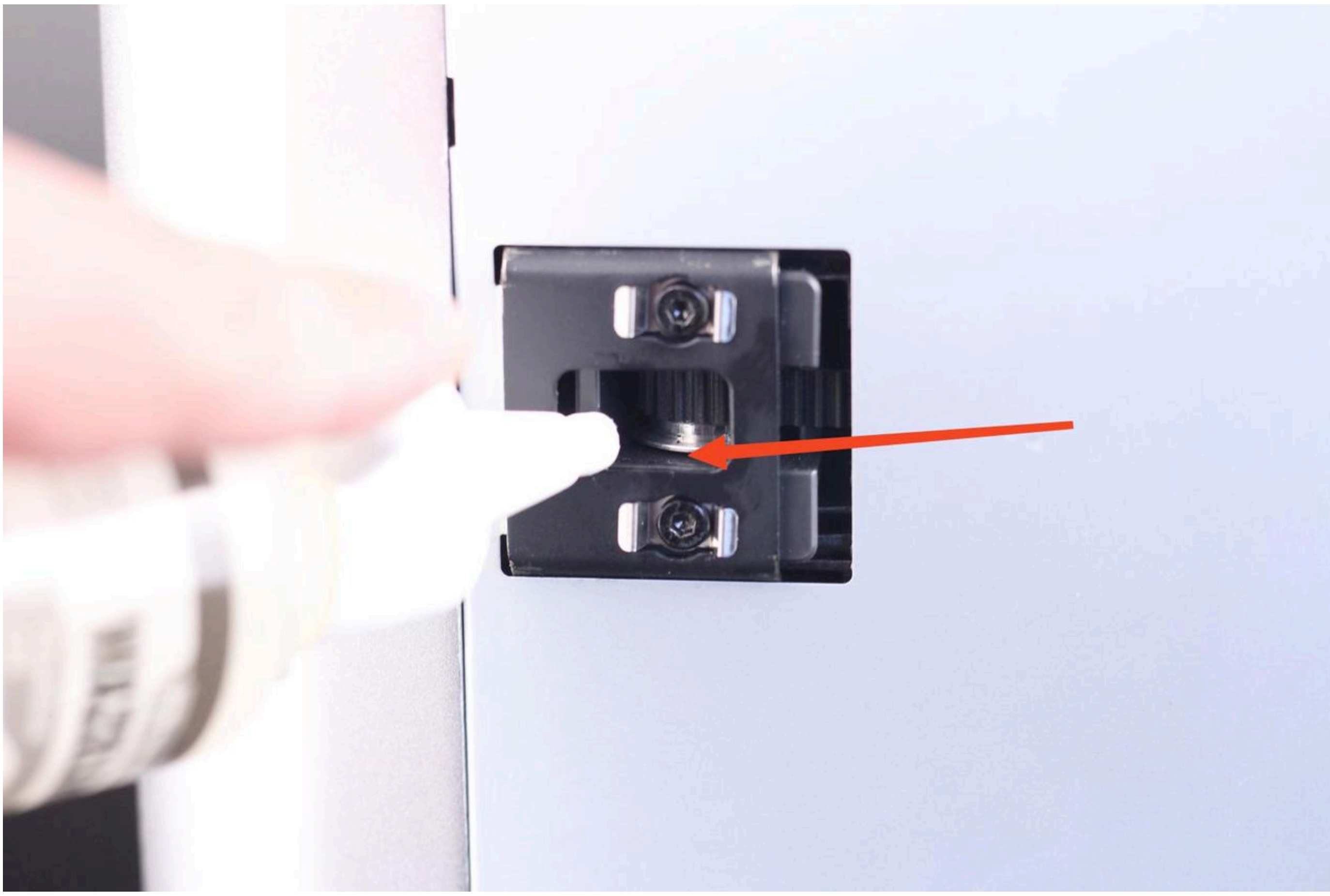
The silicone sock should be replaced if wear signs are present, or if the silicone sock no longer remains securely attached to the hotend.

How to do it

Simply remove the old silicone sock (when nozzle is at room temperature) from the hotend, and install a new one which can be purchased here .

Idler pulley

There are multiple idler pulleys installed on the printer. These idlers have sealed bearings inside which don't require greasing, but we still recommend adding a bit of lubricating oil between the idler flange and the plastic holder to avoid any squeaking noises. DO NOT add grease to the belts as this might cause the belts to slip or layer shift issues.



When to do it

When squeaking noises are present during printing, or during print head movement.

How to do it

A small amount of lubricating oil should be applied on top and on the bottom of the idler, in the location between the idler flange and the plastic holder, if needed. Avoid adding lubrication oil to the idlers if no squeaking noises are present, to avoid any dust buildup.

You can refer to using Super Lube 52004 Synthetic Lightweight Oil for better performance.

Chamber Camera (Optional)

The chamber camera lens should be cleaned regularly, to ensure a clear view.

When to do it

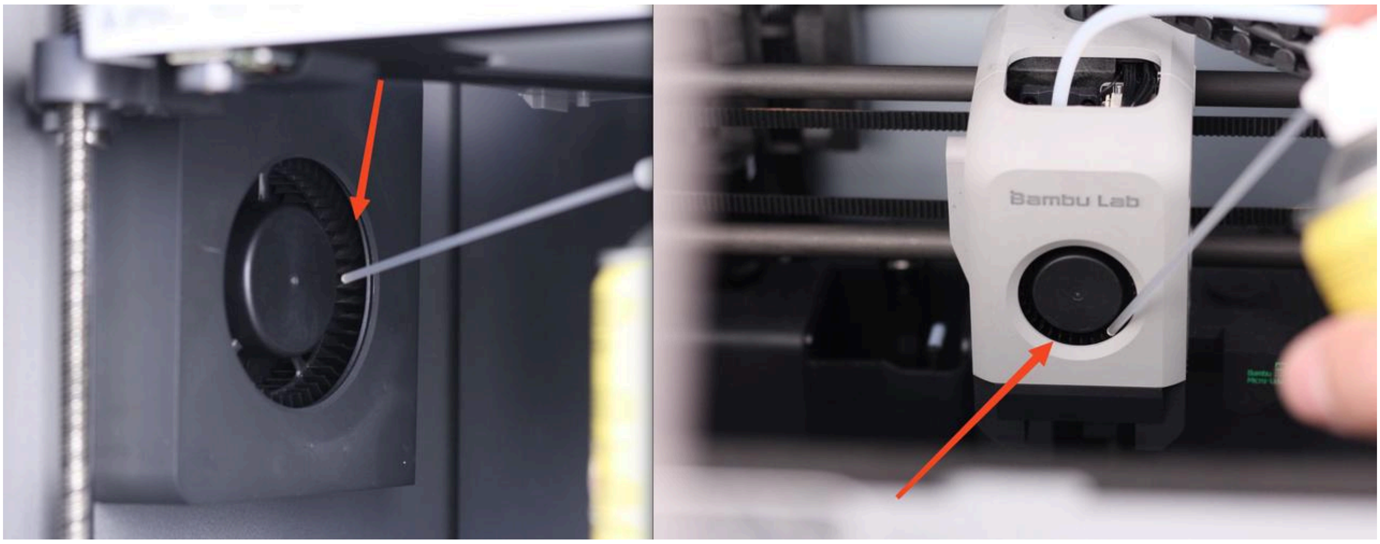
We recommend cleaning the camera lens when the video is blurred or visibly dirty. If printing ABS, we recommend cleaning it every week.

How to do it

Using a microfiber cloth and some isopropyl alcohol, gently rub the chamber camera. A q-tip can also be used as it allows the user to reach the camera easier.

Part cooling fans

There are two fans that should be regularly checked for any dust and debris build-up, the auxiliary fan and the hotend fan. They should be regularly checked for debris or dust buildup to ensure smooth operation and long life.



When to do it

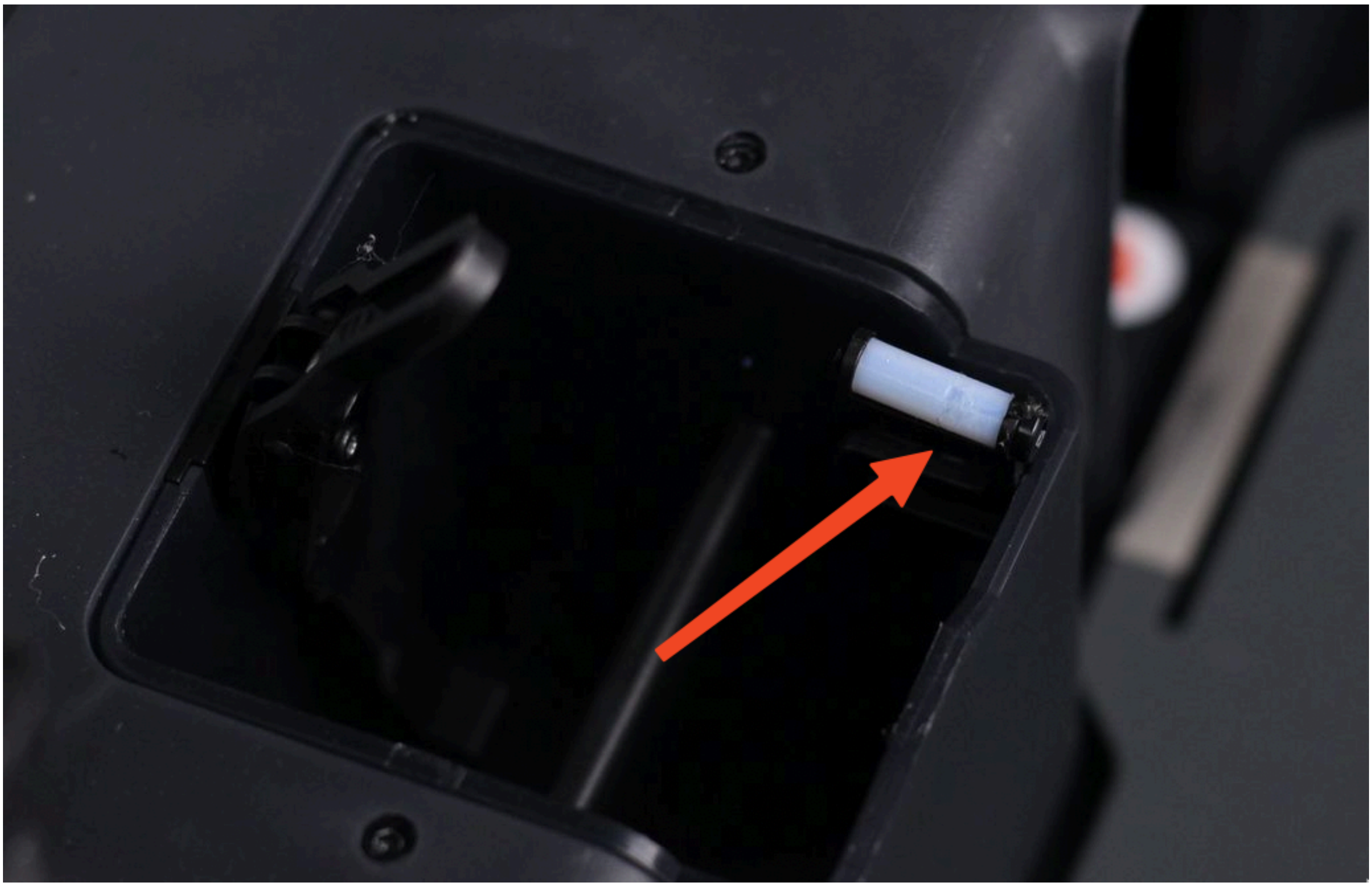
We recommend checking the fans every week to clean any debris or dust that might have built up around the blades.

How to do it

With the printer off, we recommend using a can of compressed air. While keeping the fan blades in place, use the compressed air to blow air over the blades and clean any dust or debris.

Nozzle wiper

The nozzle wiper is an important part that needs to be checked from time to time, to ensure it is undamaged, and the cleaning process works well.



When to do it

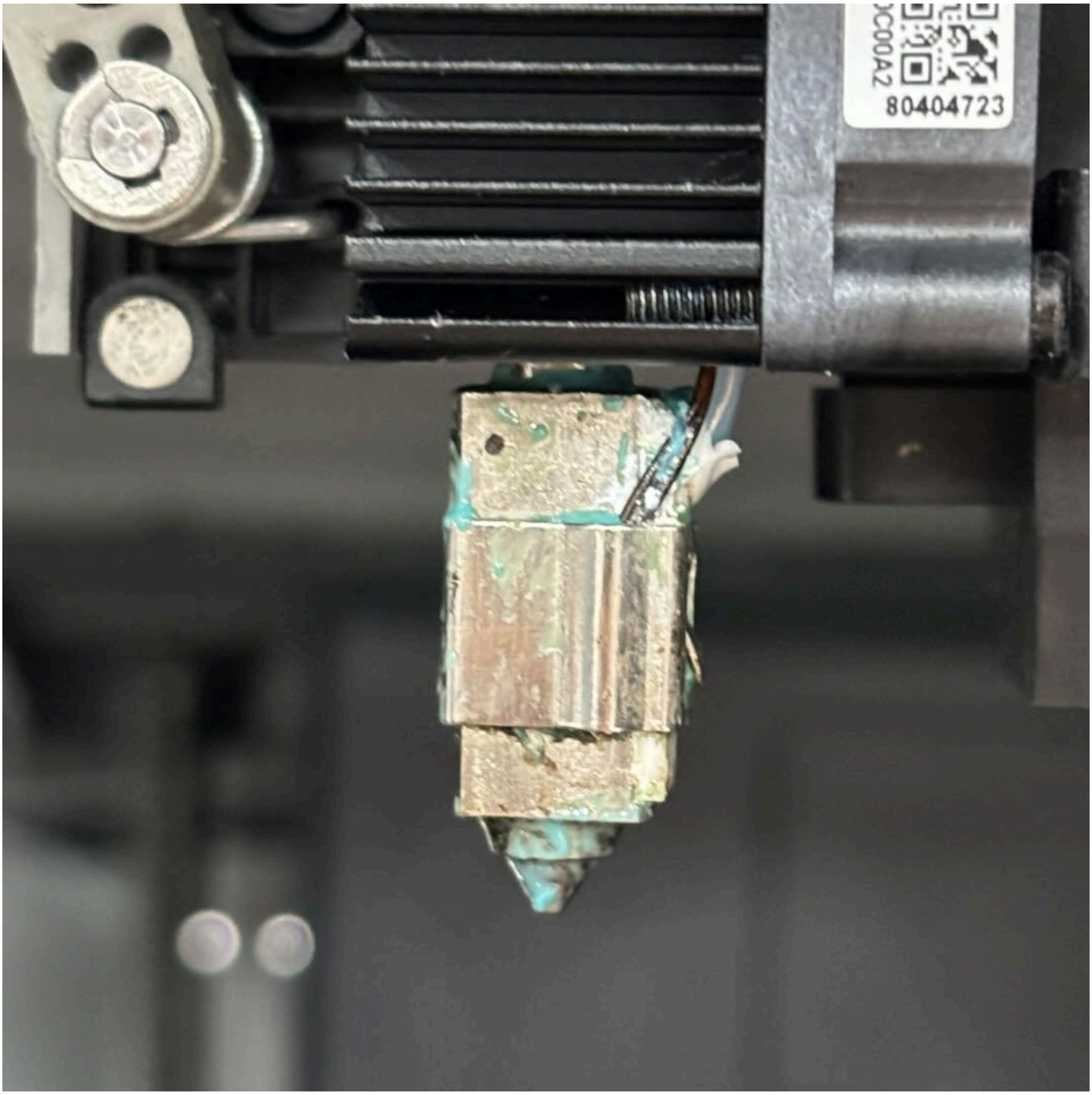
The nozzle wiper needs to be checked before starting any print, to ensure it is free of any filament debris and the PTFE side is not damaged. The wiper should also stay in a horizontal position for proper operation

How to do it

If the nozzle wiper is damaged, we recommend replacing it with a spare which can be purchased here .

Nozzle

After prolonged use, the nozzle may accumulate residue or dirt on its surface and inner walls. Regular cleaning and maintenance are necessary to prevent printing issues and prolong nozzle lifespan.



When to do it

If the nozzle surface is dirty or there is under-extrusion during printing, we recommend nozzle maintenance. Especially when printing carbon fiber materials, regular maintenance is essential to ensure extrusion stability.

How to do it

- Remove the extruder front cover, disassemble the hotend silicone sock, heat the nozzle to 200°C.
- Then, wearing heat-resistant gloves, clean the nozzle surface with tissue, cloth, or tweezers.
- Refer to the video guide below for cold pulling, clearing residue debris from the hotend and inside the nozzle.

How to perform a cold pull with PLA to clean the n
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PTFE tube

The P1 series uses PTFE tubes for guiding the filament from the AMS or external spool to the extruder. PTFE tube is quite resilient but there are some cases where it needs to be replaced.

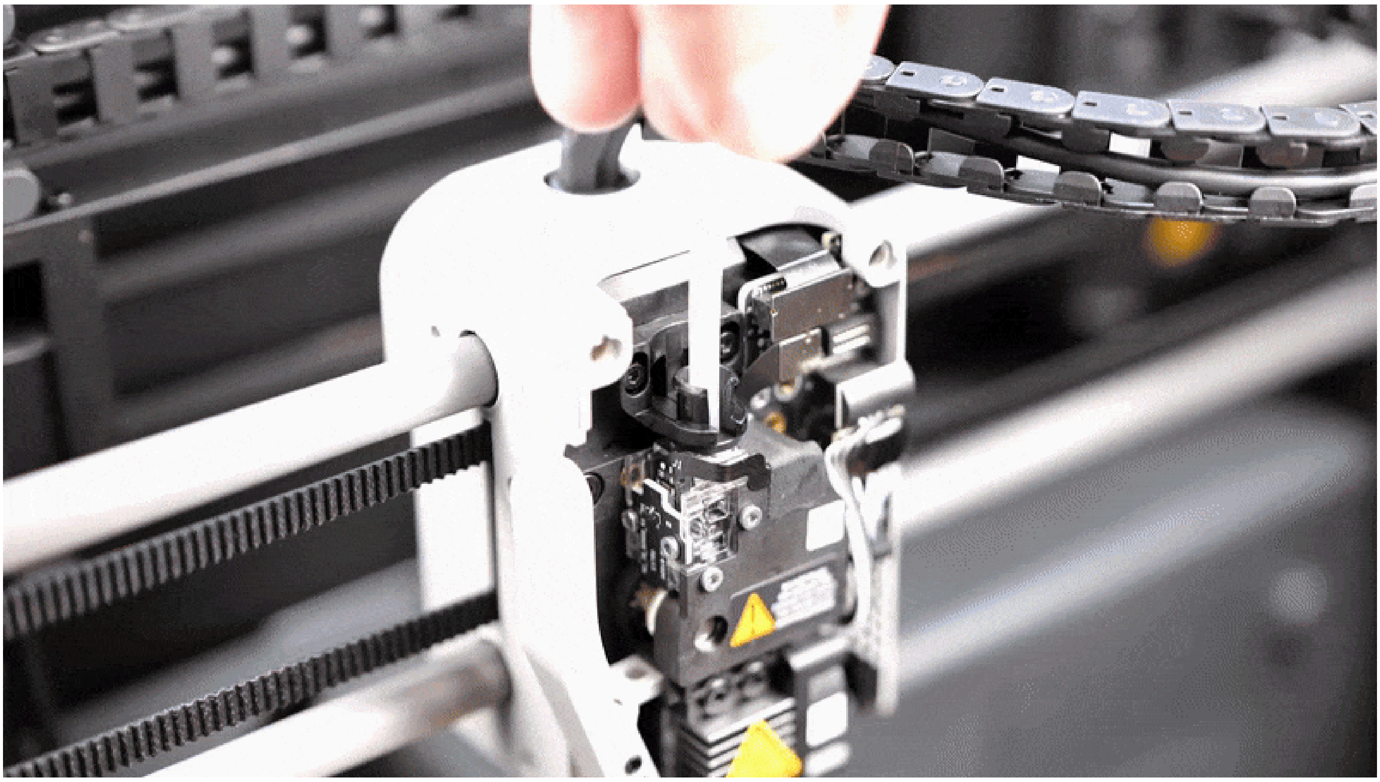
When to do it

The PTFE tube should be replaced when clear signs of wear are shown on it. Most of the time, the inside of the PTFE tube can wear out quickly when printing abrasive filaments, so we recommend checking after printing about 5 rolls of abrasive materials.

When printing regular filaments, we recommend checking the tubes after about 10 rolls of material.

How to do it

To release the PTFE tube, simply use a hex wrench to press down on the tube coupler, as shown below. While the coupler is pressed down, the tube can be easily pulled out. When doing this operation, be careful not to damage the black ribbon cable of the filament sensor.

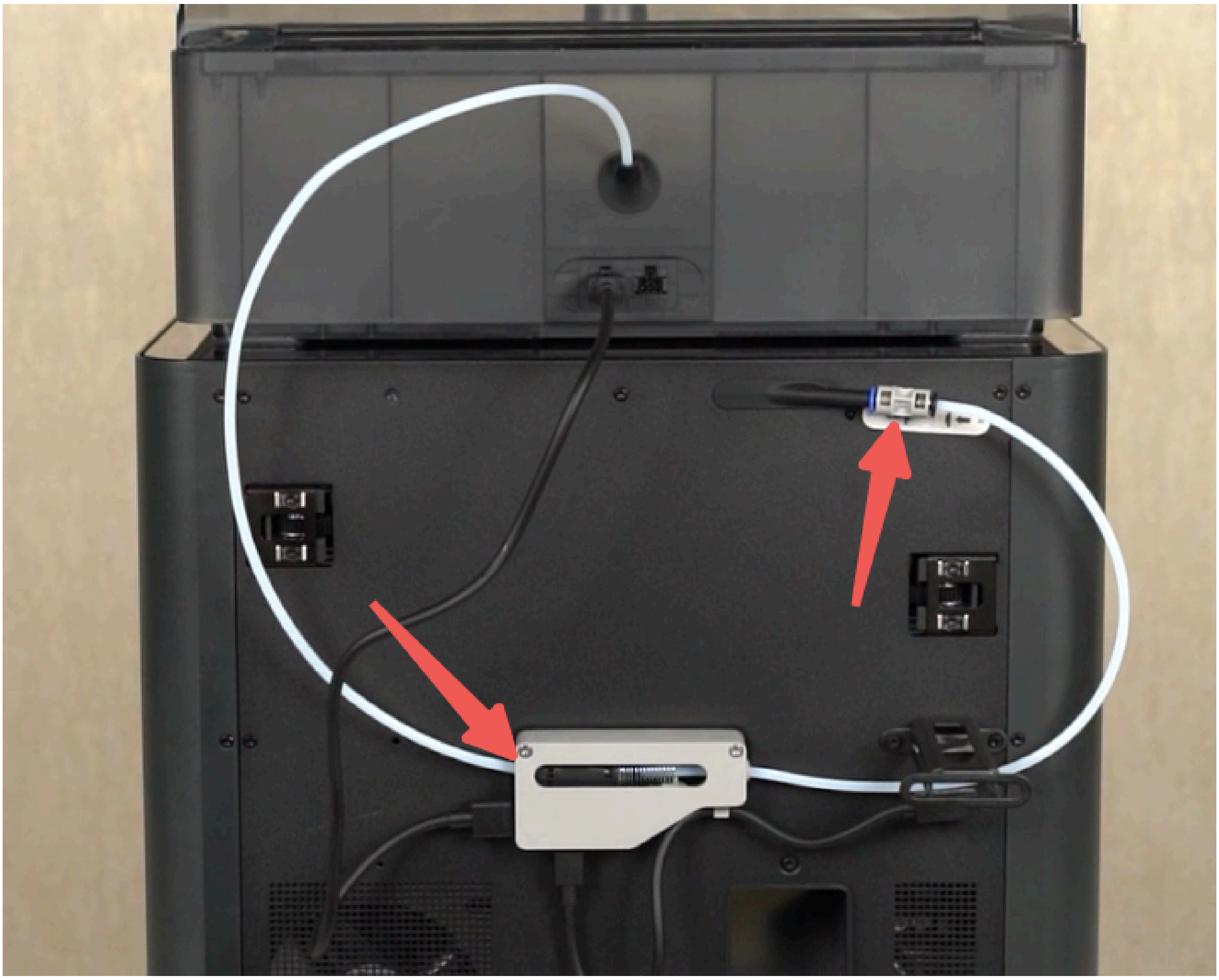


It's worth noting that you can also press on the black plastic bracket to release the tube, but you might need to loosen the two screws holding it in place before doing this operation.

PTFE Tube Connector

Issues with the PTFE couplers (which connect two PTFE tubes together) are usually solved by removing the PTFE tubing to inspect both the tubing and couplers for damage. New PTFE couplers may be purchased here .

The Filament Buffer and AMS Hub (for connecting more than one AMS to a printer) also have coupler-like PTFE connectors where PTFE tubes are inserted. If one of these couplers is damaged, generally the Buffer or Hub must be replaced. The Filament Buffer is available here and the AMS Hub is available here .



Community Contributions

[X and P Series Bambu Lab Recommended Maintenance Video](#) (By [3D_Print_Stuff](#))

This video describes some of the maintenance methods and tools for the X and P series printers.

End Notes

We hope this detailed guide has been helpful and informative.

If this guide does not solve your problem, *please submit a* *support ticket* . We will answer your questions and provide assistance.

If you have any suggestions or feedback on this Wiki, please leave a message in the comment area.

Thank you for your support and attention!